

implement quantum-resistant solutions. The importance of such assessments cannot be overstated, as proactive preparation is key to mitigating risks and maintaining operational resilience. Effective readiness assessments enable security leaders to construct roadmaps for migration, prioritise critical assets, and align business objectives with emerging technological demands.

Threat detection and attack modelling using knowledge graphs

Threat detection and attack modelling in the age of quantum computing require innovative strategies that go beyond traditional security approaches. One such advancement is the use of knowledge graphs, which provide a sophisticated means to map out and understand the intricate web of relationships between entities, events, and vulnerabilities within an organisation's digital environment. Knowledge graphs offer a structured framework, enabling security teams to integrate and analyse vast amounts of data from multiple sources. This interconnected representation allows for the identification of nuanced patterns and hidden correlations that might otherwise remain undetected, especially those associated with quantum-enabled threats.

By leveraging knowledge graphs, organisations can simulate potential attack vectors, including those that exploit the unique capabilities of quantum technology. These models help security professionals anticipate how quantum adversaries might target critical assets, exposing gaps that need to be addressed before they can be exploited. The dynamic nature of attack modelling using knowledge graphs supports continuous refinement as new threats emerge and as the organisational landscape evolves. As a result, security teams gain a holistic perspective that enhances their situational awareness, enabling them to respond more effectively to emerging risks. In essence, the adoption of

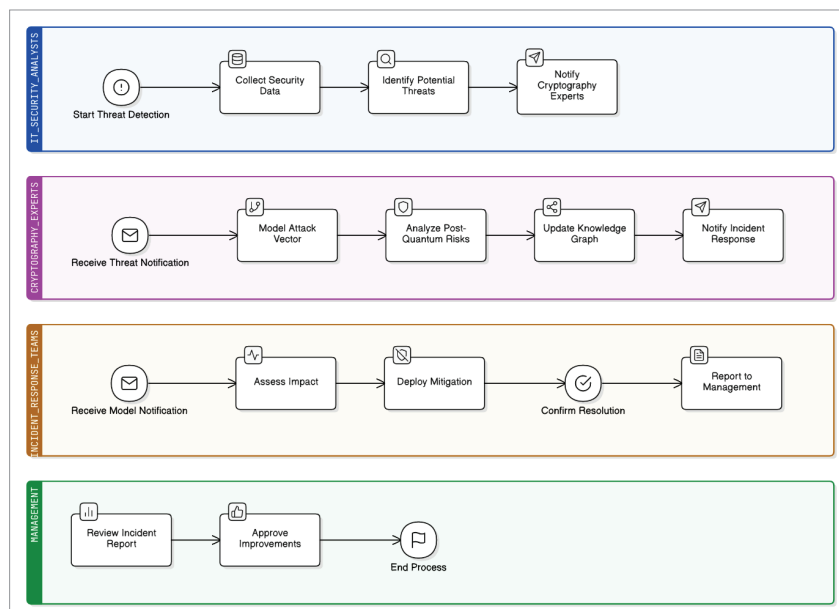


Figure 2: Business process diagram of threat detection and attack modelling

knowledge graph-based methodologies elevates the organisation's ability to detect threats early, model complex attack scenarios, and reinforce the overall security posture in preparation for the quantum era.

Continuous monitoring for quantum threats and vulnerabilities

Quantum continuous monitoring is driven primarily by the 'harvest-now, decrypt-later' cyberthreat – adversaries collect encrypted data today and decrypt it in the future when quantum systems are mature enough to do so. Thus, monitoring needs to be threat-informed for quantum capability signals and cryptography-informed for tracking wherever breakable or weak cryptography is still in use.

Quantum threat intelligence and horizon tracking: A revamped programme regularly fetches quantum-aligned and ecosystem readiness-related indicators: vendor roadmaps, error-correction achievements, logical qubit forays, and standardisation updates. This stream feeds a Quantum Risk Register which is reviewed as vulnerability exposure.

Cryptographic asset discovery and 'Y2Q' telemetry: Organisations need constant visibility of where RSA/ECC, weak parameter sets and non-agile crypto are deployed. This is done by Crypto Bill of Materials (CBOM) and automated crypto scanners for code, containers, firmware, PKI, and SaaS configurations.

Hybrid deployment and monitoring that is safe to rollback: Hybrid key exchange (classical + PQC KEM) is about the only thing that's standard while migrating to de-risk adoption. Monitoring must validate:

- Proper negotiation (no silent downgrade), matching sets of parameters, and library versions.
- Behaviour and failure scenarios (handshake latency, fragmentation, retransmissions) with respect to identifying operational regressions that could also be exploited by attackers.

Handling vulnerabilities in PQC implementations:

Continuous monitoring should include:

- Side-channel exposure analysis (timing/cache), RNG monitoring, constant-time-testing, and dependency CVE analysis in PQC-